

Microeconomics

1. A firm's profit function is

$$\pi(Q) = -Q^3 + 9Q^2 + 48Q - 100$$

where Q is the firm's output.

- (a) Find the critical point(s) of the profit function.
- (b) Use the second-order condition (SOC) to determine whether the relevant critical point is a maximum or minimum.
- (c) Calculate the firm's maximum profit.

2. Consider a market with the following inverse demand and inverse supply functions:

$$\text{Demand: } P = 120 - Q^2$$

$$\text{Supply: } P = 24 + Q^2 - 4Q$$

where P is price and Q is quantity.

- (a) Find the market equilibrium price and quantity.
- (b) Using differentiation, derive the slope of the demand curve and the slope of the supply curve.
- (c) Evaluate the slopes of both the demand and supply curves at the two mathematical intersection points. Determine whether each curve is upward-sloping or downward-sloping at each point.
- (d) Sketch the demand and supply curves. Clearly indicate both intersection points and identify the economically relevant market equilibrium.

3. A competitive market has the following characteristics:

Consumers demand 20 units when the price is zero. For every 1-unit increase in price, quantity demanded decreases by 0.5 unit.

Producers are willing to supply 2 units when the price is zero. For every 1-unit increase in price, quantity supplied increases by 0.25 unit.

Let P denote price and Q denote quantity.

(a) Derive the demand and supply functions in the form $Q=f(P)$

(b) Rewrite both functions in inverse form $P=f(Q)$, and find the market equilibrium price and quantity.

(c) Using differentiation, determine the slopes of the inverse demand and inverse supply curves. Explain the economic meaning of the signs of the derivatives.

(d) Suppose the government imposes a per-unit tax of 12 on producers. Derive the new inverse supply curve and determine the new market equilibrium.

(e) Sketch the original demand and supply curves and show how the tax changes the market equilibrium.

4. Consider the market for coffee. For each of the following independent situations, determine the effect on the equilibrium price and equilibrium quantity and illustrate your answer using a demand-and-supply diagram. Assume all other factors remain constant.

(a) A widely publicized medical study reports that regular coffee consumption may provide significant health benefits.

(b) A severe drought substantially reduces the production of coffee beans.

(c) Household incomes increase significantly, and coffee is a normal good. At the same time, technological improvements reduce the cost of coffee production.

(d) The price of tea, which is a substitute for coffee, decreases significantly. At the same time, wages paid to workers on coffee plantations increase.